

Microeconomics I

Class 1 — Preferences

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Universitat de Barcelona · Master in Economics

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Slide 1 Title

Class 1 opens Part I of Microeconomics I with three building blocks: the consumption set (MWG 2.C), binary relations and their properties, and rational preferences (MWG 1.B).

Everything later in Part I, from indifference curves to demand, starts from the two axioms defined at the end of today.

Who I am

- Studied in Pilsen, Kaohsiung, Barcelona and New York
- Mathematics first, then economics — PhD, New York University, 2019
- Across the Atlantic by sailboat, twice
- Now: the economics of AI safety and governance

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Slide 2 Who I am

A short introduction: mathematics first, then economics, a PhD from New York University, and current work on the economics of AI safety and governance.

Contact details at the bottom; office hours are on slide 11.

Plan for today

10:00	Why any of this matters
10:20	Diagnostic quiz — not graded
10:30	Part I: the nine sessions, the grade
10:40	The consumption set
11:00	Break
11:10	Answers to the exercise; quiz results
11:20	Binary relations and their properties
11:40	Rational preferences: $\succsim$, $\succ$, $\sim$

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Microeconomics I · Class 1

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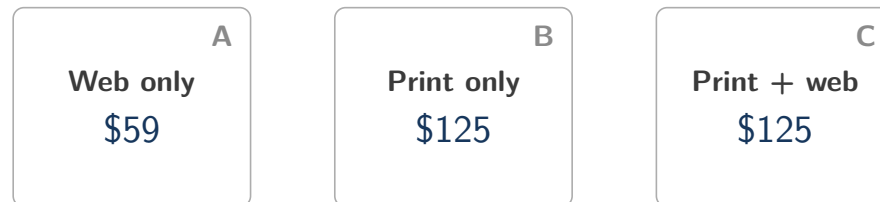
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Slide 3 Plan for today

The morning at a glance. Two hooks and a short diagnostic quiz first, then the syllabus and the grade, then the theory in three parts: the consumption set before the break, binary relations and rational preferences after it.

Straight after the break: the answers to the pair exercise and the quiz results.

The Economist's subscription offer



Hands up: A, B or C?

The offer as used in Ariely (2008), ch. 1.

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Microeconomics I · Class 1

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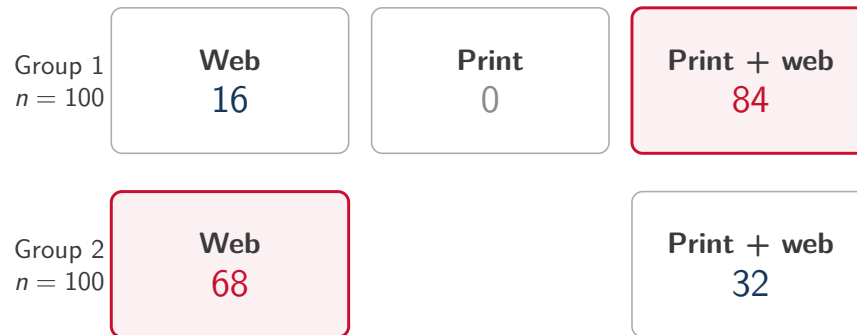
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Slide 4 The Economist's subscription offer

A real subscription menu: web only \$59, print only \$125, print and web \$125. Vote before turning the page.

Hardly anyone picks B, since C gives strictly more for the same price. The question is why B is on the menu at all.

An option nobody chose changed the choice



Another 100 students, same prices, middle option removed. Did the 84 make a **mistake**?

Ariely (2008), ch. 1; Huber, Payne & Puto (1982).

Slide 5 An option nobody chose changed the choice

Ariely's classroom experiment (2008, ch. 1). With three options, 16 of 100 students chose web, 0 print, 84 print and web. Another 100 students saw the same two prices without the print-only option: 68 chose web, 32 print and web. An option nobody chose moved the majority from the \$59 option to the \$125 one.

Caveat: two different groups and aggregate counts, not one person choosing twice. The original laboratory finding is Huber, Payne and Puto (1982), the "asymmetric dominance" effect.

Whether the 84 made a mistake, and what "consistent" would even mean, is left open until slide 35, once rational preferences are defined.

The cost of living, measured wrong by 1.1 points a year

1.1 pp

per year: how much the US CPI overstated the rise in the cost of living

0.4 pp of it: consumers substitute when prices move

Compounded, it moves pensions and tax brackets. The fix: session 6.

Boskin Commission, report to the US Senate Finance Committee (1996).

Slide 6 The cost of living, measured wrong

The Boskin Commission (1996), reporting to the US Senate Finance Committee, estimated that the US consumer price index overstated the yearly rise in the cost of living by about 1.1 percentage points. About 0.4 percentage points of that is substitution bias: the index held the basket fixed while consumers moved away from goods that got dearer.

Pensions and tax brackets are indexed to the CPI, so the error compounds into real money. The right object is the expenditure function; the fix comes in session 6.

What consumer theory lets you do

- Predict how demand reacts to prices and taxes
- Put a euro value on a policy: who gains, who loses
- Test rationality on purchase data, no utility needed
- Price risk: insurance, portfolios, lotteries

BLP (1995) · Hausman (1981) · Samuelson (1938), Afriat (1967) · von Neumann & Morgenstern (1944).

Slide 7 What consumer theory lets you do

Four things the theory of Part I is used for, each with a landmark paper. Demand systems predict how demand reacts to prices and taxes (Berry, Levinsohn and Pakes 1995 for cars; the same logic prices a sugar tax). Welfare measures turn “who gains, who loses” into euros (Hausman 1981).

Revealed preference tests rationality on purchase data without ever writing a utility function (Samuelson 1938, Afriat 1967); not on our syllabus. Choice under risk, insurance and portfolios (von Neumann and Morgenstern 1944), is sessions 8 and 9.

Diagnostic quiz

- Eight questions — **not graded**
- Shows me what to explain and what to skip
- Your name on the sheet, circle the letters
- About ten minutes, then tear off the strip and hand it in

Not seen it before? Leave it blank, do not guess.

Slide 8 Diagnostic quiz

Eight questions, not graded, about ten minutes. It shows what needs explaining and what can be skipped, nothing else.

Circle the letters rather than shading. If you have never seen a topic, leave it blank instead of guessing. Tear off the answer strip and hand in only the strip; keep the questions.

Results after the break, slides 19 and 20.

Part I in nine sessions

		MWG
1	Tue 29 Sep	Consumption set, binary relations, preferences
2	Wed 30 Sep	Monotonicity, convexity, indifference curves
3	Tue 6 Oct	Utility representation
4	Wed 7 Oct	Utility maximisation, Walrasian demand
5	Tue 13 Oct	Indirect utility, price and wealth effects
6	Wed 14 Oct	Expenditure minimisation, Hicksian demand
7	Tue 20 Oct	Slutsky equation; welfare measures
8	Wed 21 Oct	Uncertainty I: expected utility, risk aversion
9	Tue 27 Oct	Uncertainty II: Arrow–Pratt, dominance; review

Wed 28 Oct Part I exam

MWG · Muñoz-Garcia, *Advanced Microeconomic Theory*, ch. 1, 2, 5 (gentler).

Slide 9 Part I in nine sessions

The nine sessions of Part I with the MWG sections behind each: preferences and utility (sessions 1 to 3), demand (4 to 7), choice under uncertainty (8 and 9). The Part I exam is in class on Wednesday 28 October.

MWG is the text. Muñoz-Garcia, *Advanced Microeconomic Theory*, covers the same material more gently, with worked examples.

How Part I is graded

Part I exam, Wed 28 Oct, in class	40 %
Two problem sets, groups of 2–5	5 %
Short quizzes in class, on paper	5 %

Groups of 2–5: sign-up sheet **tomorrow**. Quizzes weekly, closed book, lowest dropped.

Part II: Olga Chiappinelli, from November. Pass: 40 % in every component, 50 % overall.

Slide 10 How Part I is graded

Part I is half the course grade: 40 % for the exam on 28 October, 5 % for two problem sets done in groups of 2 to 5, 5 % for short quizzes in class, on paper, closed book, weekly, with the lowest dropped.

The sign-up sheet for groups goes round tomorrow; come with your group. Passing by continuous assessment needs 40 % in every component and 50 % overall. Part II is taught by Olga Chiappinelli from November.

Practicalities

- Course page: pkocourek.com/teaching/micro-1
- Office hours: Wednesdays 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Problem set 1 out Tue 6 Oct, due Wed 14 Oct
- AI: none in quizzes and exams; in problem sets, declare it

Slide 11 Practicalities

Slides and reading: pkocourek.com/teaching/micro-1.

Office hours: Wednesdays 12:00–13:00, after class, up to 20 minutes per student; booking by e-mail gives priority.

Problem set 1 is handed out on Tuesday 6 October and due on Wednesday 14 October.

AI tools: none in quizzes and exams; in problem sets they may be used but must be declared.

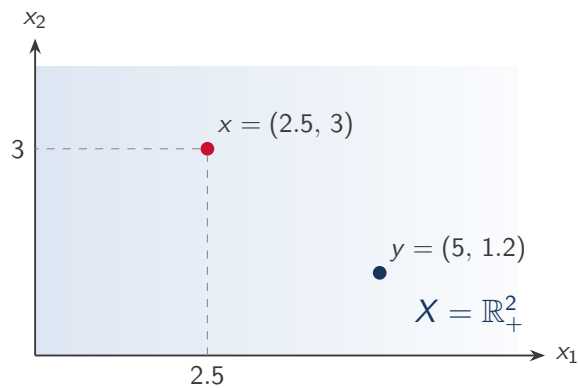
The consumption set

MWG 2.C

Slide 12 The consumption set

Part 1 of the theory: the set of bundles a consumer could physically consume. MWG 2.C.

Two goods, one point



A bundle is a vector. X is what can be consumed; affordability comes later.

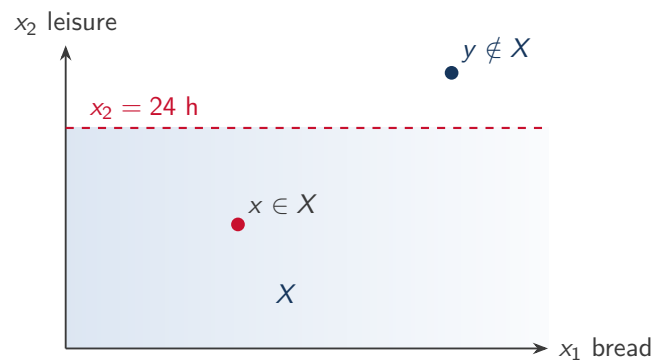
Slide 13 Two goods, one point

A bundle is a vector: $x = (2.5, 3)$ is 2.5 units of good 1 and 3 units of good 2, one point in the plane.

The consumption set X is the set of all bundles the consumer could physically consume. Here $X = \mathbb{R}_+^2$, the whole non-negative quadrant (shaded).

Prices and wealth are not part of X : it describes what is possible, not what is affordable; affordability is the budget set, MWG 2.D, session 4. Quantities are per period, say kilograms of bread per week or hours of leisure per day.

Physical limits cut the set



$X = \{x \in \mathbb{R}_+^2 : x_2 \leq 24\}$. A day has 24 hours.

After MWG 2.C, redrawn.

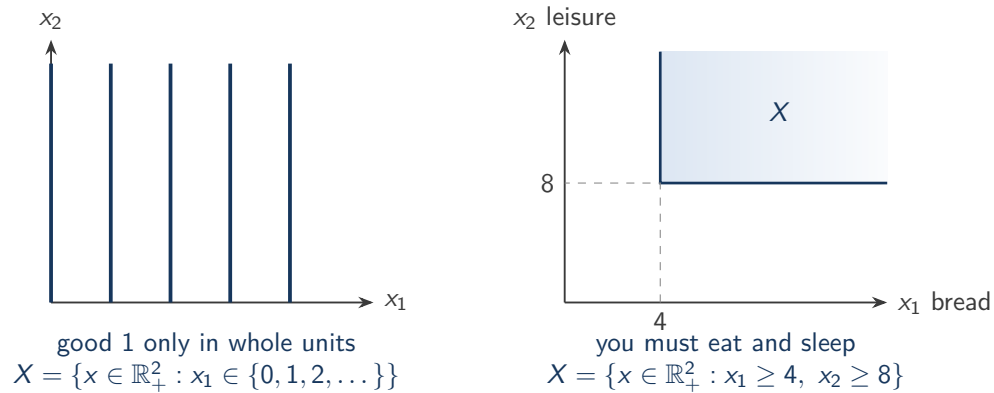
Slide 14 Physical limits cut the set

Physical limits shrink X . If good 2 is hours of leisure per day, nothing above $x_2 = 24$ can be consumed, so $X = \{x \in \mathbb{R}_+^2 : x_2 \leq 24\}$, the shaded strip below the dashed line.

Read the notation as “all x in $\mathbb{R}_+^2$ such that $x_2 \leq 24$ ”: the braces build a set, the colon reads “such that”, the condition selects. The same grammar defines every set in this course, including the budget set.

The blue point y lies above the line: it is not a bundle anyone can consume, so $y \notin X$.

Indivisible units, survival



After MWG 2.C, redrawn.

Slide 15 Indivisible units, survival

Two more cuts, after MWG 2.C. Left: good 1 comes only in whole units, so X is a set of vertical lines at $x_1 = 0, 1, 2, \dots$; any amount of good 2 goes with each. Right: survival needs at least 4 units of bread and 8 hours of leisure, so X is the region to the upper right of the corner $(4, 8)$.

From chapter 3 on, MWG simply takes $X = \mathbb{R}_+^L$; these pictures are the reasons that is a simplification. X is normally assumed convex, which is what the exercise on the next slide tests.

In pairs: sketch X

1. Both goods sold only in whole units
2. Bread in Barcelona (x_1) or bread in Madrid (x_2) — you can be in one place
3. Leisure between 8 and 24 hours a day, bread any amount

Three minutes. Which of the three sets are **convex**?

Slide 16 In pairs: sketch X

Try it, three minutes: sketch the consumption set for each of the three situations, with x_1 on the horizontal axis and x_2 on the vertical.

Then decide which of the three sets are convex. A set is convex if the straight segment between any two of its points stays inside the set: for $x, y \in X$ and $\lambda \in [0, 1]$, $\lambda x + (1 - \lambda)y \in X$.

Answers on slide 18, after the break.

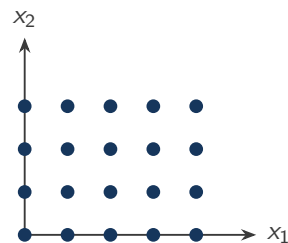
Ten minutes; back at 11:10.

Break

Back at 11:10

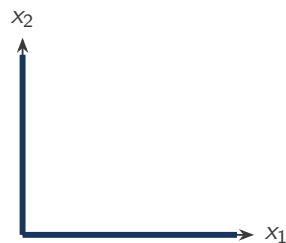
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Answers: sketch X



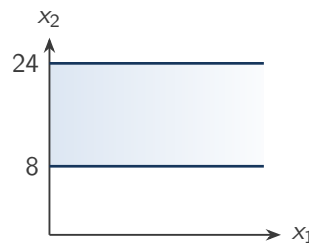
1. $x_1, x_2 \in \{0, 1, 2, \dots\}$

not convex



2. $x_1 = 0$ or $x_2 = 0$

not convex



3. $8 \leq x_2 \leq 24$

convex

Convex: the segment between any two points of X stays in X . Only the third set passes.

Slide 18 Answers: sketch X

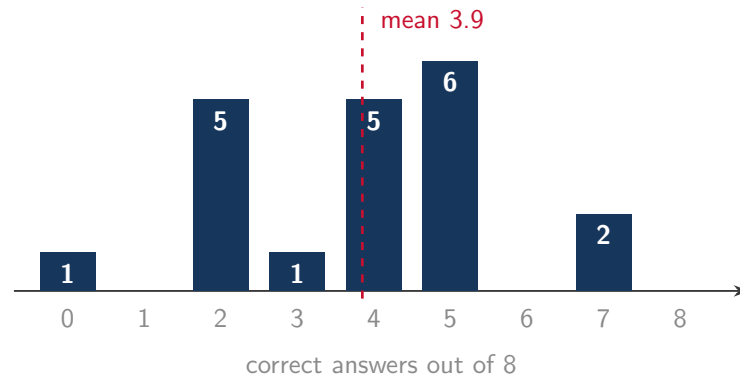
1. Both goods in whole units: X is the lattice of integer points. Not convex: the segment from $(0, 0)$ to $(1, 1)$ passes through $(0.5, 0.5)$, which is not in X .

2. Bread in Barcelona or bread in Madrid: you are in one place, so $x_1 = 0$ or $x_2 = 0$, and X is the union of the two axes. Not convex: the segment from $(1, 0)$ to $(0, 1)$ leaves the set.

3. Leisure between 8 and 24 hours, bread in any amount: the horizontal strip $8 \leq x_2 \leq 24$. Convex: the segment between any two points of the strip stays in it.

Convexity of X belongs to MWG 2.C. Convexity of preferences is a different notion and comes tomorrow.

Diagnostic quiz: scores



20 strips. Median 4. Not graded.

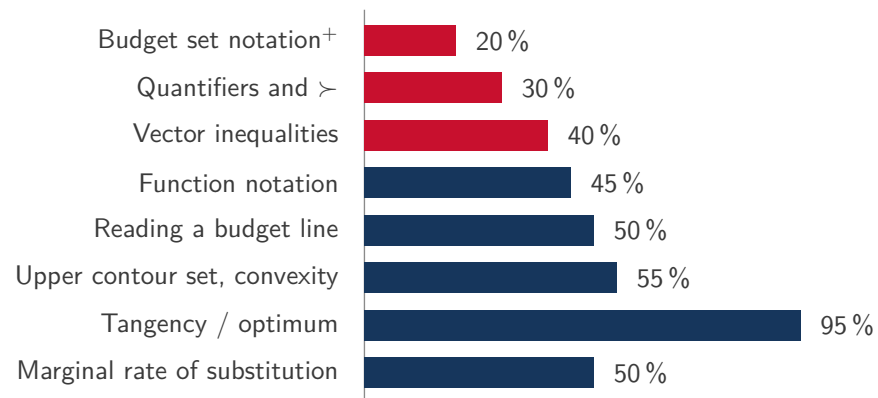
A question with several correct options counts only if the whole set is right.

Slide 19 Diagnostic quiz: scores

Scores of the diagnostic quiz: 20 strips, correct answers out of 8, mean 3.9, median 4. A question with several correct options counted only if the whole set was right.

Not graded. The purpose is to see where class time is best spent; the next slide shows where.

Diagnostic quiz: share right, by question



$n = 20$. ⁺ select all that apply: the whole set must be right.

Slide 20 Diagnostic quiz: share right, by question

Share of correct answers by question, $n = 20$. The economics was fine: 95 % had the tangency question. Three pieces of notation were not (red): set-builder notation for the budget set (20 %, a select-all question, so the whole set had to be right), the order of the quantifiers $\forall$ and $\exists$ with $\succ$ (30 %), and the vector inequalities $\geq$ versus $\gg$ (40 %).

Set-builder notation is used today on slides 14 and 15; the same grammar defines every set in the course. The quantifiers and vector inequalities return tomorrow, where monotonicity needs them.

Binary relations

and their properties

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Slide 21 Binary relations

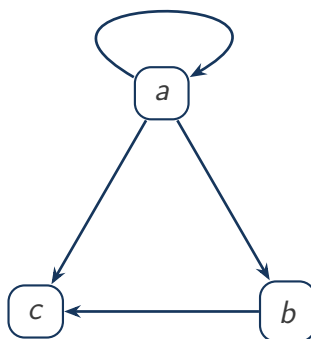
Part 2: binary relations and their properties. A preference will be one of these, nothing more.

Definition: binary relation

A **binary relation** R on X is a set $R \subset X \times X$.

Write $x R y$ for $(x, y) \in R$.

$\geq$ on $\mathbb{R}$ · “is taller than” · $\subset$ on sets



$$R = \{(a, a), (a, b), (b, c), (a, c)\}$$

Slide 22 Definition: binary relation

A binary relation R on a set X is a set of ordered pairs of elements of X , that is, a subset of $X \times X$. Writing $x R y$ means the pair (x, y) belongs to R .

Picture: $X = \{a, b, c\}$ and $R = \{(a, a), (a, b), (b, c), (a, c)\}$. Each arrow is one pair, and the loop at a is the pair (a, a) . The pair (b, a) is not in R , so $b R a$ is false.

Everyday examples: $\geq$ on the real numbers, “is taller than” on the people in the room, $\subset$ on sets. A preference is a relation of exactly this kind, with X the consumption set.

Properties of a relation, I

complete: for all x, y : $x R y$ or $y R x$ $\geq$ on $\mathbb{R}$

reflexive: for all x : $x R x$ “has the same birthday as”

transitive: $x R y$ and $y R z \Rightarrow x R z$ “is an ancestor of”

Complete $\Rightarrow$ reflexive. Take $y = x$.

Slide 23 Properties of a relation, I

Three properties. Complete: any two elements are related in at least one direction; for $\geq$ on $\mathbb{R}$, of any two numbers one is at least the other. Reflexive: every element is related to itself. Transitive: whenever $x R y$ and $y R z$, also $x R z$; chains close.

Complete implies reflexive: apply the definition of completeness with $y = x$.

“Has the same birthday as” is reflexive but not complete: two people with different birthdays are related neither way. “Is an ancestor of” is transitive but neither reflexive nor complete.

Properties of a relation, II

symmetric: $x R y \Rightarrow y R x$ “is a sibling of”

asymmetric: $x R y \Rightarrow \text{not } y R x$ “is taller than”

antisymmetric: $x R y$ and $y R x \Rightarrow x = y$ $\geq$ on $\mathbb{R}$

Asymmetric $\Rightarrow$ antisymmetric, and **irreflexive:** never $x R x$.

Slide 24 Properties of a relation, II

Three more. Symmetric: whenever $x R y$, also $y R x$ (siblings).

Asymmetric: whenever $x R y$, never $y R x$ (“is taller than”).

Antisymmetric: both directions hold only for equal elements; for $\geq$ on $\mathbb{R}$, $a \geq b$ and $b \geq a$ force $a = b$.

Asymmetric implies irreflexive (take $y = x$: $x R x$ would contradict itself) and antisymmetric (the premise of antisymmetry never occurs).

A relation can be neither symmetric nor asymmetric: $\geq$ on $\mathbb{R}$ runs both ways between 3 and 3, and only one way between 3 and 2.

In pairs: which properties hold?

	complete	reflexive	symmetric	transitive
$\geq$ on $\mathbb{R}$	✓	✓	✗	✓
$>$ on $\mathbb{R}$	✗	✗	✗	✓
"is a sibling of"	✗	✗	✓	✗
$x \geq y$ in both coordinates, on $\mathbb{R}_+^2$	✗	✓	✗	✓
$x_1 + x_2 \geq y_1 + y_2$, on $\mathbb{R}_+^2$	✓	✓	✗	✓

Siblings: transitivity is a claim for *all* a, b, c , including $c = a$.
Then $a R b$ and $b R a$ would force $a R a$ — and no one is their own sibling.

Slide 25 In pairs: which properties hold?

Try it before reading the ticks (four minutes): for each of the five relations decide complete, reflexive, symmetric, transitive.

$>$ on $\mathbb{R}$: $3 > 3$ fails both ways, so neither complete nor reflexive; asymmetric, hence not symmetric. "Is a sibling of": symmetric, not reflexive, and not transitive. Transitivity is a claim for all a, b, c , including $c = a$: with $c = a$, $a R b$ and $b R a$ would require $a R a$, and no one is their own sibling. Take the "for all" in the definition seriously. Symmetric plus transitive forces $a R a$ only for elements related to something, so the two together still do not give reflexive.

Coordinatewise $\geq$: reflexive and transitive, not complete (next slide), a partial order. $x_1 + x_2 \geq y_1 + y_2$: complete, reflexive, transitive: the first rational preference of the course, before the word is defined.

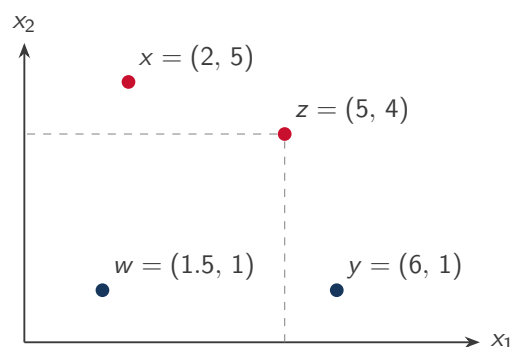
Slide 26 Coordinatewise $\geq$ is not complete

Coordinatewise $\geq$ ranks two bundles only when one has at least as much of both goods: $z = (5, 4)$ has more of both than $w = (1.5, 1)$, so $z \geq w$.

Between $x = (2, 5)$ and $y = (6, 1)$ it says nothing: x has more of good 2, y more of good 1. "More of everything" is the easy case; economics starts where the vector order stops.

A preference must rank x against y anyway. That requirement is completeness, the first axiom on slide 29.

Coordinatewise $\geq$ is not complete



$z \geq w$, but x and y are not ranked. A preference must rank them anyway.

Part 3: the preference relation and the two axioms that make it rational. MWG 1.B.

Rational preferences

MWG 1.B

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The preference relation

$\succsim$ is a binary relation on X : $x \succsim y$ “ x is at least as good as y ”

$x \succ y \iff x \succsim y \text{ and not } y \succsim x$ strict preference

$x \sim y \iff x \succsim y \text{ and } y \succsim x$ indifference

MWG 1.B.

Slide 28 The preference relation

MWG 1.B. The preference relation $\succsim$ is a binary relation on the consumption set X ; $x \succsim y$ reads “ x is at least as good as y ”.

Two relations are derived from it. Strict preference $x \succ y$: x is at least as good as y , but y is not at least as good as x . Indifference $x \sim y$: each is at least as good as the other.

Indifference is not the same as “not strictly preferred”: if x is not strictly preferred to y , it may be that y is strictly preferred to x .

Definition: rational preferences

$\succsim$ is **rational** if it is

complete: for all $x, y \in X$: $x \succsim y$ or $y \succsim x$

transitive: for all $x, y, z \in X$: $x \succsim y$ and $y \succsim z \Rightarrow x \succsim z$

Completeness: you can always rank two bundles — even ones you have never faced.

MWG Definition 1.B.1.

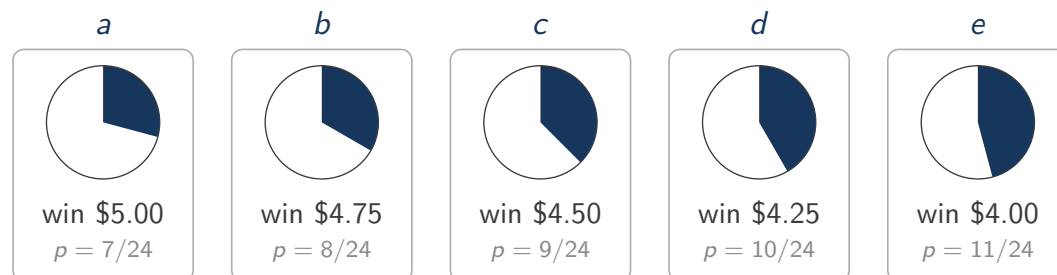
Slide 29 Definition: rational preferences

MWG Definition 1.B.1. $\succsim$ is rational if it is complete (for any two bundles x, y in X , x is at least as good as y , or y is at least as good as x , or both) and transitive (if x is at least as good as y and y at least as good as z , then x is at least as good as z).

These are the two properties from the exercise table. Rationality here is consistency of rankings, nothing about being clever or selfish.

Completeness is most doubtful for choices far from experience (say, a job in a city you have never seen). Transitivity is doubtful when differences are small: the next two slides.

Five gambles: which do you take?



a or b? b or c? ... a or e?

Shaded sector: the chance of winning. Subjects saw the discs, not the fractions.

Tversky (1969), *Psychological Review* 76, Experiment I, Table 1.

Slide 30 Five gambles: which do you take?

Tversky (1969), Experiment I. Five gambles: a wins \$5.00 with probability $7/24$, b \$4.75 with $8/24$, c \$4.50 with $9/24$, d \$4.25 with $10/24$, e \$4.00 with $11/24$. The prize falls and the odds rise from a to e ; the expected value rises.

Subjects saw only the shaded discs, not the fractions. Try it pair by pair: a or b ? b or c ? c or d ? d or e ? Then a or e .

Transitivity: what people did

$$a \succ b, \quad b \succ c, \quad c \succ d, \quad d \succ e$$

and yet $e \succ a$

8 of 18 students screened showed this pattern: small probability gaps ignored, big ones not.

Tversky (1969), *Psychological Review* 76(1), 31–48.

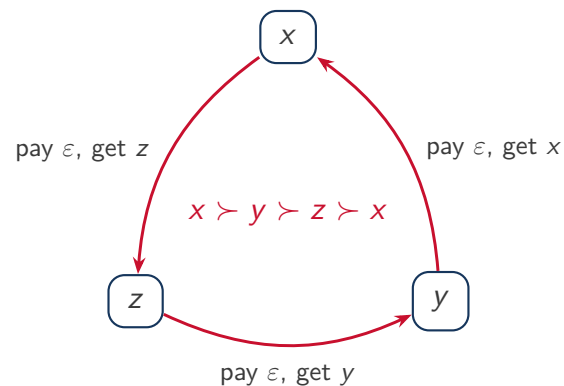
Slide 31 Transitivity: what people did

The pattern Tversky found. Neighbouring gambles differ by only $1/24$ in probability, a gap the discs hide, so the higher prize wins each adjacent comparison: $a \succ b \succ c \succ d \succ e$. Between a and e the gap of $4/24$ is visible, and $e \succ a$: a cycle.

8 of the 18 students screened showed this pattern and went on to the main experiment. The mechanism: small differences in one dimension are ignored, large ones are not.

Transitivity is an assumption about people, and it can fail in predictable ways.

Why cycles cost you: the money pump



Same object, 3ε poorer, ready to go round again.

Davidson, McKinsey & Suppes (1955), *Philosophy of Science* 22.

Slide 32 Why cycles cost you: the money pump

Why a cycle is costly. Suppose $x \succ y \succ z \succ x$. Someone holding z pays a small amount ε to swap to y , ε more to swap to x , and ε more to swap back to z : the same object as at the start, 3ε poorer, and willing to go round again.

The money pump (Davidson, McKinsey and Suppes 1955) is an argument, not evidence: it explains why we would expect learning or markets to punish cycles, not that people are in fact transitive.

What rationality gives you for free

If $\succsim$ is rational, then

- (i) $\succ$ is irreflexive and transitive hence asymmetric
- (ii) $\sim$ is reflexive, symmetric and transitive an equivalence relation
- (iii) $x \succ y \succsim z \Rightarrow x \succ z$

Prove (iii) in two lines. Which of the two axioms does it use?

MWG Proposition 1.B.1.

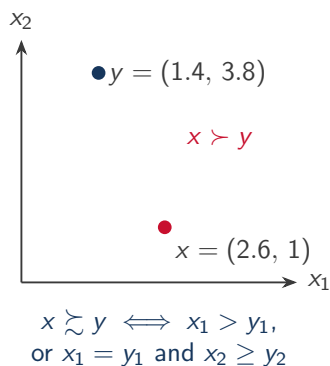
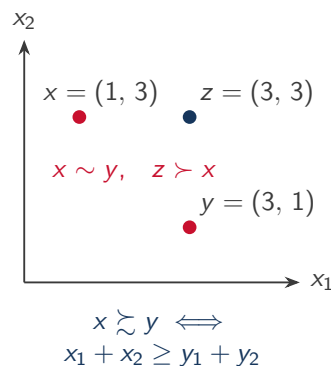
Slide 33 What rationality gives you for free

MWG Proposition 1.B.1. If $\succsim$ is rational, then (i) strict preference is irreflexive and transitive, hence also asymmetric ($x \succ y$ and $y \succ x$ would give $x \succ x$); (ii) indifference is reflexive, symmetric and transitive, an equivalence relation, so X splits into indifference classes (tomorrow: indifference curves); (iii) $x \succ y$ and $y \succsim z$ together give $x \succ z$.

Proof of (iii): from $x \succ y$ and $y \succsim z$, transitivity gives $x \succ z$. Suppose also $z \succsim x$; then $y \succsim z$ and $z \succsim x$ give $y \succsim x$, contradicting $x \succ y$. So $x \succ z$. Only transitivity is used.

Completeness is what makes $\sim$ reflexive: with $y = x$ it gives $x \succsim x$.

Two preferences on $\mathbb{R}_+^2$ — rational?



Two minutes with your neighbour: complete? transitive? What do $\sim$ -classes look like?

Slide 34 Two preferences on $\mathbb{R}_+^2$ — rational?

Two minutes with your neighbour: is each preference complete? transitive? What do its indifference classes look like?

Left: $x \succsim y$ whenever $x_1 + x_2 \geq y_1 + y_2$. Complete and transitive because $\geq$ on the sums is: rational. In the picture $x = (1, 3) \sim y = (3, 1)$ and $z = (3, 3) \succ x$; the indifference classes are the lines $x_1 + x_2 = c$.

Right, lexicographic: compare good 1; only if tied, good 2. One of the two steps settles any pair (complete), and it is transitive case by case: rational, even though $x = (2.6, 1) \succ y = (1.4, 3.8)$ and every indifference class is a single point. Remember this one: on Tuesday 6 October it shows that a rational preference need not have a utility function.

Back to the decoy

{ Web, Print + web }

most chose **Web**

{ Web, Print, Print + web }

most chose **Print + web**

If each student picks the $\succsim$ -best item, Web vs Print + web is settled before any menu is shown.
Adding Print, which nobody chose, cannot flip it — yet Web went from 68 to 16.

Groups 1 and 2 of the reveal slide. Ariely (2008), ch. 1.

Slide 35 Back to the decoy

Premise: each student picks the $\succsim$ -best item on the menu. A rational $\succsim$ ranks Web against Print + web before any menu is shown, and adding Print, which nobody chose, cannot reverse that ranking. Two random samples of 100 should then split about the same way, not 68 % Web in one and 16 % in the other.

The only escapes: most people are exactly indifferent, or the menu itself changes the preference (the decoy makes Print + web look like a bargain). That is the point of writing $\succsim$ down: it does not depend on the menu. Caveat: group data, not one person choosing twice.

Today in one line



Tomorrow: what indifference curves look like when $\succsim$ is monotone and convex.

Read MWG 3.B (or Muñoz-Garcia ch. 1) for Wednesday. Bring your group of 2–5.

Slide 36 Today in one line

Today in one line: a consumption set $X \subset \mathbb{R}_+^2$; a binary relation $R \subset X \times X$ on it; and the two axioms, complete and transitive, that make it a rational preference $\succsim$.

Tomorrow, Wednesday 30 September: what indifference curves look like when $\succsim$ is monotone and convex (MWG 3.B).

Reading for Wednesday: MWG 3.B, or Muñoz-Garcia chapter 1. Bring your group of 2 to 5 for the sign-up sheet.